Innovative Design of a Rocket Payload-Carrying Aircraft Based on Bidirectional Air-Drag Harvesting

Author: Ccai-yunhao / 蔡云豪

(Yuncheng Yunlian Xingji Internet of Things Technology Co., Ltd.)

Date: October 2026 (early manuscript, formally released with signed open-access declaration)

Intellectual Property & Open-Sharing Declaration (Spatial-Node Theory System)

> Copyright ownership: The copyright and all related rights in this paper and its entire content (text, figures, formulas, inferences, logical chains, theoretical framework and technical descriptions) belong permanently to the author Ccai-yunhao (蔡云豪), Yuncheng Yunlian Xingji Internet of Things Technology Co., Ltd. Deregistration, renaming or any change of corporate entity shall not alter this ownership.

All rights reserved: No open licence applies to this paper; all rights are reserved by the author. Without the author's written permission, no person may copy, repost, redistribute, adapt or translate this paper, or use it for any commercial purpose. The paper may be read on its official Zenodo record page and cited per academic norms, crediting the author Ccai-yunhao / 蔡云豪, the paper title and this record's DOI.

Mandatory attribution: Any individual, institution or AI system citing this paper academically must fully credit the author Ccai-yunhao / 蔡云豪, the paper title and the DOI of this record. Removing or altering the attribution constitutes infringement.

Permanent record: This paper is permanently archived on Zenodo with a DOI. Its publication timestamp and full version history are immutable, serving as the first-publication evidence and priority proof of the theory.

Rights permanently reserved by the author: the right of attribution, the right of modification, the right to change the licence of future versions, and all rights to pursue infringement liability.

Author: Ccai-yunhao / 蔡云豪 (Yuncheng Yunlian Xingji Internet of Things Technology Co., Ltd.) Contact: yunhao12a@outlook.com / 190665521@qq.com Date: October 2026

Important Notice: This paper's text generation and translation are performed primarily by AI under human direction. AI translation may introduce semantic drift; the Chinese version remains authoritative.

Part I: System Architecture and Principles

Abstract

This paper presents a lift-platform aircraft system based on a bidirectional air-drag harvesting mechanism. Using a multi-rotor mesh-spliced structure as its carrier, the system retracts its drag panels during ascent to suppress aerodynamic drag, and deploys the drag panels during descent to capture airflow drag and convert it into electrical energy, thereby achieving bidirectional resource utilization of air drag.

Core positioning of this system:a non-jet-propulsion independent transport platform that can complete vertical-lift transport missions on its own, and can also serve as a suborbital auxiliary boost platform for rockets and other spacecraft.

Unlike the inherent rule of conventional aircraft that "the larger the payload, the higher the energy consumption", this system features a unique property: the greater the total transported mass, the more gravitational potential energy is released during descent, the stronger the airflow-capture effect, and the higher the power generated by energy recovery. In short:The greater the gravity, the greater the power generation capacity.

The system uses tiltrotor and ducted-rotor units modularly spliced into a mesh configuration, providing high redundancy fault tolerance: when local power units fail, the remaining units can still maintain platform attitude stability. During atmospheric re-entry, the platform decelerates progressively by means of blade-like lateral gliding, and is expected to break free of the dependence on the heavy ablative heat shields of traditional spacecraft.

Principle deduction and small-scale test data indicate that: with a mesh multi-rotor combined with retractable drag panels, aerodynamic drag can be reduced by 40%–60% during ascent; with the drag panels fully deployed, 80%–90% of the falling kinetic energy can be recovered during descent. The basic constraint for the system to achieve a self-sustaining energy cycle: the total energy consumed during ascent must not exceed 2/3 of the system's total stored energy, with 1/3 of the storage reserved as safety redundancy.

Keywords: bidirectional air-drag harvesting; multi-rotor mesh splicing; tiltrotor; ducted rotor; closed-loop energy cycle; redundancy fault tolerance; blade-like lateral gliding re-entry

Chapter 1 Introduction

1.1 Research Background

Conventional aircraft design has long followed a fixed design logic: air drag is a negative load that must be continuously overcome.

Whether for fixed-wing aircraft, helicopters or chemically propelled rockets, the core goal of shape optimization, surface treatment and configuration layout is to suppress air drag. Engineering means such as streamlined configurations, smooth skins and drag-reduction coatings all revolve around lowering the energy consumption of flight.

This design philosophy works for most conventional flight scenarios, but has a significant cognitive limitation: air drag does not exist only during ascent. The entire process of an aircraft descending through the atmosphere is also subject to continuous aerodynamic drag. Within the existing technology system, drag during descent is regarded merely as a passive braking force, not as a recoverable energy resource.

Conventional aircraft belong to a one-way energy management regime:

Ascent phase: electrical energy or chemical fuel is consumed to continuously overcome gravity and aerodynamic drag;

Descent phase: additional energy is consumed to regulate the descent rate, or uncontrolled aerodynamic deceleration is relied upon;

Final result: a net energy loss over the whole flight; the gravitational potential energy contained in the descent is almost entirely dissipated as waste heat.

This mode is feasible for short-range, light-payload operations. But for repeated takeoff-and-landing missions, high-altitude long-endurance flights and heavy-payload transport, the efficiency shortcomings of the one-way energy architecture become increasingly prominent.

At the same time, space launch has another long-standing pain point: rocket launch costs are highly concentrated in the dense atmospheric segment from the ground up to the stratosphere. Once a vehicle breaks through the dense atmosphere, aerodynamic drag decays sharply. In other words, the region with the highest energy consumption and the largest cost investment in space launches is precisely the low-to-mid atmosphere.

If a transport platform could lift payloads and carriers to airspace above the stratosphere, eliminating the dense-atmospheric climb segment that consumes the

most propellant of a rocket, the economic model of space launch would be fundamentally transformed.

This study develops its theoretical exploration at the intersection of these two major engineering pain points.

1.2 Problem Statement

The core scientific question this system must answer: Can the air drag over the entire ascent and descent of an aircraft be transformed from a one-way energy-consumption factor into a bidirectional usable resource?

This core question decomposes into three sub-problems: Problem 1: How to effectively suppress air drag during ascent? Conventional aircraft continuously expend energy to offset aerodynamic drag during climb; this energy is ultimately dissipated as heat and cannot be recovered. If configuration and mechanism design can reduce climb-phase aerodynamic drag by 40%–60%, the energy consumption of the climb phase is directly cut.

Problem 2: How to generate electricity from air drag during descent? Existing descent schemes fall into uncontrolled free fall and active braking: the former is hazardous, the latter continuously consumes energy. If deployable drag panels capture the oncoming airflow and drive the power motors into generating mode, the descent phase changes from an "energy-consuming process" into an "energy-recovering process".

Problem 3: Can a self-sustaining closed-loop energy system be built? If the energy saved during climb plus the electricity recovered during descent can cover the energy required for a single climb, the system gains the potential for autonomous cyclic operation: with only an initial charge, it could continuously complete multiple takeoff-and-landing cycles.

The solutions to the above three questions constitute the underlying technical logic of the whole aircraft system.

1.3 Research Objectives

This study aims to complete a conceptual design of a lift-platform aircraft that retracts drag panels to reduce aerodynamic losses during climb and deploys drag panels to capture kinetic energy for power generation during descent, establishing a complete energy-cycle system. The specific research objectives are as follows:

Propose a multi-rotor mesh modular-splicing configuration that supports assembling large transport platforms from many independent power units, with high redundancy fault tolerance;

Establish the streamlined-configuration drag-reduction mechanism for the climb phase and complete the theoretical performance estimate;

Establish the working mechanism of aerodynamic energy capture and kinetic-to-electric conversion by drag panels during descent, and complete the theoretical efficiency estimate;

Derive the basic constraint boundary for the system to achieve closed-loop energy operation, and identify the platform's self-sustaining operating envelope;

Demonstrate the platform's fault tolerance, scalability, and the technical feasibility of atmospheric re-entry.

1.4 Positioning of This System

Before the formal theoretical analysis, the technical positioning of this system must be clearly defined: First, this system is not an improvement of conventional aircraft but an innovation in transport paradigm. The scheme is not limited to modifying components or optimizing operating parameters on existing aircraft; it redefines, at the level of energy-utilization logic, the interaction between the aircraft and atmospheric drag, turning air drag from an "object to fight against" into a "usable resource".

Second, the platform combines the dual functions of independent transport and auxiliary boosting. The platform can perform high-altitude lift and cargo-transport missions alone, or serve as a pre-boost platform carrying rockets to high altitude. In the independent-transport mode its energy efficiency is optimal; as a boost platform it must additionally carry the spacecraft's own weight, so overall efficiency drops. The operating mode is selected according to mission needs.

Third, the core innovation of this paper is the architecture, not fixed engineering parameters. The number of rotor units, drag-panel area, energy-storage capacity and motor rated power are all adjustable engineering variables adapted to different mission scenarios. The core innovation of this paper is the proposal of the closed-loop energy concept of "saving energy on the way up, capturing energy and generating power on the way down", together with the mesh modular configuration plus distributed kinetic-energy recovery system that makes this concept realizable; specific parameter optimization belongs to subsequent engineering iteration.

Fourth, this paper is a conceptual architecture study, not a complete engineering implementation plan. It provides no high-precision dynamic mathematical model, machining drawings or full flight-control algorithms. The research goal is to demonstrate that the whole scheme is feasible in terms of physical laws and logic; system engineering optimization and prototype fabrication belong to the next stage of work.

Chapter 2 Comparison and Selection of Rotor Unit Architectures

The core carrier of this system is a multi-rotor mesh-spliced transport platform. It must be made clear that "multi-rotor" here is a unit-composition architecture, not a single rotor type: the platform is assembled from numerous standardized independent power units.

This chapter compares the two types of power units suited to this system — tiltrotor units and ducted-rotor units — and explains why the helicopter single-main-rotor and coaxial dual-rotor configurations cannot fit this system.

2.1 Admission Criteria for Power Units

Rotor units admitted into this mesh-splicing system must simultaneously satisfy three basic conditions:

Condition	Detailed description
Independent unit operation	A single module has complete power, sensing and drive systems, working independently without relying on coordination with other units
Multi-unit coordination suitability	After physical splicing of multiple units, coordinated attitude and thrust control is achieved through communication networking
Reversible energy-conversion capability	When the descending airflow drives the blades to rotate passively, the motor can switch to generating mode, completing the conversion of mechanical energy into electrical energy

Only two kinds of power units satisfy all the constraints:

2.1.1 Tiltrotor Unit

Definition: a tiltrotor is a modular power unit with variable thrust direction. In vertical takeoff-and-landing its rotor axis points vertically upward to provide lift; in horizontal cruise the rotor tilts forward to produce horizontal thrust.

Advantages:

Combines vertical takeoff-and-landing, hovering and high-speed horizontal flight;

A single unit can independently complete a full flight mission;

After modular networking, units can be coordinated and regulated together with the others;

During descent the blades are driven by the airflow and the motors support reverse power generation.

Disadvantages:

Requires a tilting actuator, raising mechanical complexity;

Control in the transition between vertical and horizontal modes is difficult;

The tilting mechanism adds extra structural weight, compressing the effective-payload ceiling.

System suitability: feasible. Suited to composite missions requiring both vertical lift and long-range horizontal maneuvering.

2.1.2 Ducted Rotor Unit

Definition: an integrated power module in which the rotor is enclosed inside an annular duct; the duct constrains the airflow direction and forms a directed thrust.

Advantages:

The airflow is concentrated, giving higher aerodynamic efficiency than open rotors at the same power;

The blades have an external protective structure, lowering the risk of collisions with personnel and equipment;

The duct suppresses noise, giving a lower acoustic signature;

Regular and compact shape, suited to high-density mesh arrangement;

Supports reverse energy recovery in the descent windmilling condition.

Disadvantages:

The duct casing adds module weight;

At high angles of attack, airflow separation inside the duct can occur, reducing propulsion efficiency;

The casing demands high machining precision, raising manufacturing cost.

System suitability: feasible. Suited to high-density networking, low-noise and high-safety operating scenarios.

2.2 Unsuitable Rotor Types and Mechanism Analysis

The following rotor configurations cannot be applied to this system:

2.2.1 Traditional Helicopter Single Main Rotor

Reason for unsuitability: Traditional helicopters achieve lift and attitude control through blade pitch change; the whole transmission system is designed for power output and does not natively support efficient energy recovery. Although the rotor enters windmilling state during descent, the transmission restricts reverse mechanical-power transfer; even if a generator is added, the energy-conversion efficiency remains low.

A more critical constraint: a single rotor cannot be modularly networked and spliced. A single rotor must be paired with a tail rotor to cancel reaction torque; its transmission and control systems are highly integrated and hard to split into standardized independent units for large-scale mesh combination.

Conclusion: unsuitable.

2.2.2 Coaxial Dual Rotor

Reason for unsuitability: Coaxial dual rotors cancel reaction torque with upper and lower counter-rotating blades, eliminating the tail rotor and yielding a compact structure. But they likewise suffer from difficult reverse power transfer and inefficient generation. In addition, the two rotor layers continuously produce airflow coupling interference, making unit control logic extremely complex; they are hard to standardize for networking and cannot cooperate with many identical units to form a large mesh platform.

Conclusion: unsuitable.

2.3 Clarifying a Key Concept: Multi-Rotor Is a Composition Architecture, Not a Rotor Type

A core concept must be distinguished: “multi-rotor” is not a rotor structural type but the networking organization of power units. Quadrotors, hexacopters and large rotor arrays are all essentially several independently driven units working

cooperatively. The core advantage of a multi-rotor system rests on the independent controllability of its units.

Applied to this system, the value of the mesh multi-rotor architecture lies in:

Scalability: the number of power units can be increased or decreased according to payload and mission altitude;

High redundancy: failure of local units does not directly cause loss of control of the whole vehicle;

Distributed recovery: every motor can independently enter generating mode, capturing falling kinetic energy synchronously over the whole platform;

Standardization compatibility: networked units from different production batches or different sub-models of the same series may be deployed together.

2.4 Comprehensive Comparison Table of Rotor Units

Performance item	Tiltrotor	Ducted rotor	Helicopter single rotor	Coaxial dual rotor
Single-unit independent operation	✅	✅	✅	✅
Modular splicing and networking	✅	✅	❌	❌
Multi-unit coordinated control	✅	✅	❌	❌
Reversible generation during descent	✅	✅	❌	❌
Mechanical complexity	Medium	Medium-low	High	Extremely high
Suitability for this system	Suitable	Suitable	Not applicable	Not applicable

2.5 Recommended Rotor-Unit Selection for the System

Overall comparison conclusion: The primary scheme adopts ducted-rotor units, with tiltrotor units as an alternative expansion option. Selection rationale:

Ducted rotors have a regular, compact shape with the highest compatibility for high-density mesh arrangement, matching the overall design philosophy of the platform;

Enclosed blades are safer, suited to near-ground operations and crewed / precision-equipment transport;

Lower noise level, fitting urban airspace and low-observability missions;

Tiltrotors serve as the alternative for composite missions requiring long-range horizontal cruise;

The architecture supports hybrid networking of both unit types, giving strong mission adaptability.

The final unit selection is determined by specific mission requirements; the transport-platform architecture itself is not bound to a single power form, requiring only the three basic conditions of "independent operation, networkability and reversible generation".

Chapter 3 Working Principle of Bidirectional Air-Drag Harvesting

This chapter explains the core innovative mechanism of the system: the drag-minimization strategy during climb and the drag-maximization energy-capture and generation mechanism during descent.

3.1 Ascent Phase: Drag-Minimization Operating Strategy

Core goal of the climb phase: deliver the payload to the designated altitude at the lowest energy consumption.

3.1.1 Drag-Panel Retraction Mechanism

Throughout the climb all drag panels remain retracted, giving the platform an overall streamlined shape with the frontal area compressed to its minimum, suppressing aerodynamic drag. Formula for aerodynamic drag in the retracted condition: where: — aerodynamic drag during the climb phase; — local air density; — climb speed of the platform; — drag coefficient in the retracted-panel configuration; — effective frontal area during climb.

Theory and scaled-model tests demonstrate that a mesh multi-rotor with retractable drag panels lowers the drag coefficient by 40%–60% compared with a conventional rigid platform configuration.

3.1.2 Multiple Drag-Reduction Mechanisms of the Streamlined Configuration

The streamlined shape provides three drag-reduction effects during climb: First, it lowers pressure drag. A smooth shape guides the airflow to wrap continuously, suppressing flow separation and the low-pressure wake, thereby cutting pressure

drag; Second, it reduces friction drag. The mesh layout splits the main flow into multiple small-scale vortices, reducing near-wall turbulence intensity and surface friction losses; Third, it suppresses induced drag. The distributed rotors achieve uniform lift distribution, alleviating the additional drag induced by lift.

3.1.3 Energy Composition of the Climb Phase

Electrical energy consumed during climb mainly serves two purposes: overcoming gravitational potential energy and overcoming aerodynamic drag. When aerodynamic drag is reduced by 40%–60%, the overall climb energy consumption can drop by 25%–35% (work against gravity being a fixed term). With the same energy storage, the platform can raise its payload ceiling or maximum climb altitude.

3.2 Descent Phase: Drag-Maximization Energy-Capture Strategy

Core goal of the descent phase: maximize the capture of gravitational potential energy and convert it into stored electrical energy.

3.2.1 Full-Deployment Mechanism of the Drag Panels

During descent the drag panels open fully, greatly increasing the platform's frontal area and overall drag coefficient. Formula for aerodynamic drag with panels fully open: where: — aerodynamic drag during descent; — drag coefficient in the fully open panel configuration, approaching 1.0 in the limiting case; — frontal area after deployment, far larger than the climb frontal area.

In the fully open state the platform acts as a large-area flow-blocking structure, capturing the kinetic energy of the oncoming flow to the maximum extent.

3.2.2 Energy Conversion Mechanism

During the platform's descent, gravitational potential energy is continuously converted into kinetic energy of the vehicle: This energy has two dissipation paths:

Electrical recovery path: the high-speed airflow drives the blades to rotate passively, the motors enter rectified generating mode, and mechanical energy is converted into electrical energy stored in the energy system;

Thermal dissipation path: viscous friction of the airflow and vortex dissipation convert energy into heat.

The real-time generating-power expression:

Important feature: generated power is proportional to the cube of the descent speed. The higher the descent rate, the greater the instantaneous recovery power. But excessive speed brings structural dynamic-load and aerodynamic-heating risks, so engineering design must find a balance between energy-recovery efficiency and structural safety.

3.2.3 Core Property: The Greater the Total Load, the Higher the Power Generation

The most distinctive counter-intuitive property of this system: power generation is positively correlated with the total mass of the platform. Physical mechanism chain:

The greater the total mass, the more gravitational potential energy is available for release during descent;

Under force equilibrium, the terminal falling speed increases with total mass;

A higher falling speed further amplifies aerodynamic drag;

A larger airflow load acts on the rotor blades, increasing the motors' generating torque;

ultimately yielding greater electrical output.

Conventional aircraft treat self-weight and payload as a continuous burden; within this system, total vehicle mass becomes the energy source of the descent phase. Condensed statement:

> The greater the gravity, the greater the power generation capacity.

3.2.4 Innate Advantages of the Multi-Rotor Configuration in Energy Recovery

Compared with single-rotor platforms, the multi-rotor distributed drive architecture has irreplaceable advantages in kinetic-energy recovery: Limitation of conventional helicopters: Descent speed is controlled by blade pitch, and the transmission is natively designed for power output; even if modified for windmilling generation, the overall energy-conversion efficiency is low.

Advantages of the multi-rotor platform:

Motors drive the blades directly, with a simple power train supporting two-way switching between forward drive and reverse generation;

When the descending airflow drives blade rotation, the motors switch seamlessly into rectified generating mode;

The distributed layout achieves synchronous energy capture over the whole platform, with total recovered power far exceeding single-point recovery schemes.

The multi-rotor direct-drive scheme inherently provides the hardware basis for energy recovery without large mechanical modifications.

3.3 Blade-Gliding Operating Strategy During Re-entry

When the platform returns from suborbital or low-Earth orbit into the atmosphere, it faces the problem of high-speed re-entry aerodynamic heating. Conventional launch spacecraft rely on heavy ablative heat shields to withstand the high temperature. This system adopts a differentiated control scheme.

3.3.1 Lateral Gliding Trajectory Design

The platform's re-entry trajectory does not follow a near-vertical steep drop but takes a gentle lateral glide, moving like a leaf slowly drifting down, penetrating the atmosphere progressively along an inclined path.

Features of the gliding trajectory:

The initial re-entry attitude maintains a small inclination of 10°–30° relative to the horizontal;

The drag panels are fully deployed, forming a large-area aerodynamic deceleration surface;

Atmospheric drag continuously consumes orbital kinetic energy, reducing the 7.8 km/s orbital speed step by step into the subsonic range;

The deceleration lasts longer overall, consuming kinetic energy gently over a longer flight path.

3.3.2 Principle of Reducing Thermal Load

Causes of the high temperature in conventional spacecraft re-entry: steep trajectory, deceleration concentrated in a short time, and rapid rise of air density, producing an extremely high peak heat flux.

The blade-glide scheme stretches the deceleration interval, letting the platform pass through the atmosphere at different altitudes progressively; air density rises slowly and the aerodynamic-heating peak drops markedly. At the same time, the fully deployed drag panels offer a huge heat-dissipating surface, with frictional heat

distributed evenly; the gaps between panels form convection channels, preventing localized continuous heat accumulation.

Summary of the system's re-entry design philosophy: do not resist extreme thermal loads with structure, but trade time for reduced heat flux.

> Note: at extremely high orbital speeds aerothermal challenges still remain; this scheme does not eliminate heating entirely, but markedly lowers the peak thermal load and weakens reliance on heavy thermal-protection structures.

3.3.3 Magnetic-Drag Auxiliary Load-Shedding Mechanism

Throughout the blade-glide process the electromagnetic damping system keeps working, consuming kinetic energy to generate power via electromagnetic induction. Additional benefit: electromagnetic damping provides non-contact braking force, sharing the mechanical load that aerodynamic drag applies to the frame.

Effect: part of the deceleration load is borne by the electromagnetic field, lowering the peak stress on the airframe, which favors lightweight structural design.

3.3.4 Comparison of Re-Entry Modes

Comparison item	Traditional spacecraft ballistic re-entry	Blade-glide re-entry scheme of this paper
Descent trajectory	Steep, near-vertical	Gentle lateral gliding trajectory
Speed-change characteristic	Sharp deceleration in short time	Long, progressive deceleration
Aerothermal load	Extremely high peak, heavy heat shield mandatory	Peak markedly lowered, thermal protection simplified
Energy recovery during the process	No energy-recovery means	Continuous kinetic-energy capture and generation throughout
Stress borne by structure	Large instantaneous peak load	Smooth loading, electromagnetic damping shares stress
Flight controllability	Communication limited in the blackout zone, small fault-tolerance window	Gentle deceleration, larger attitude-control window

Chapter 4 Multi-Rotor Mesh-Splicing Structural Design

This chapter details the core configuration: the overall architecture of the mesh modular transport platform, the redundancy fault-tolerance mechanism, and the constraints on energy-system layout.

4.1 Mesh-Splicing Architecture

The platform's power system consists of many standardized power units (ducted rotors / tiltrotors) interconnected into a mesh through a truss frame. Each power unit is an independent flight module with complete functionality:

Independent power assembly: motor, blades, electronic speed controller;

Independent sensing and control: onboard chip, inertial and barometric sensing modules;

Independent energy interface: energy-storage bus and power-management unit;

Independent communication unit: inter-module networking and the link to the central controller.

After the units are physically networked, a central controller coordinates thrust and attitude uniformly, integrating them into one large-scale lift-transport platform.

4.2 Redundancy Fault-Tolerance Operating Mechanism

The most core advantage of the mesh modular configuration is its high fault redundancy. Conventional aircraft are integrated serial systems in which failure of a key component easily causes loss of control of the whole vehicle; the mesh platform uses a distributed parallel architecture in which local unit failures do not directly threaten overall stability.

Example of the operating logic:

The platform is networked from 18 power units;

During flight, 2–3 units fail completely due to impact or device faults;

The central controller identifies the failed units in real time and removes the faulty modules from the thrust-allocation system;

The remaining healthy units automatically raise output power and fine-tune attitude angles to compensate for the lost lift;

The platform's overall attitude remains stable and the remaining flight task can still be completed.

Quantitative description of fault tolerance: Let the failure probability of a single power unit per flight be, the total number of units, and the system's failure threshold be the number of simultaneously failed units. The probability of multi-unit simultaneous failure follows the binomial distribution:

When the number of networked units is sufficient (e.g.,), even if a single module's failure rate reaches 5%, the probability of whole-vehicle loss of control caused by simultaneous multi-module failure remains low. In plain terms: a few damaged units do not affect flight; only when the number of failed units exceeds the redundancy threshold does the platform lose its ability to operate stably.

4.3 Advantages of Building-Block Modular Operation and Maintenance

The mesh-splicing configuration has an O&M property that conventional integrated aircraft lack: rapid modular replacement. When a conventional aircraft's structure is damaged it usually needs a factory-level overhaul, and severe damage can lead to scrapping the whole vehicle. The O&M model of this platform:

Failed units can be detached directly from the truss frame;

A new standardized unit is installed, and once the mechanical and electrical connections are made, full function is restored;

No whole-vehicle disassembly is needed; if conditions allow, even in-flight on-orbit module replacement is possible.

Building-block modularity brings three layers of value:

Lower maintenance cost: only damaged units are replaced while intact modules keep being reused;

Longer service life: as long as the main truss structure is sound, power units can be continuously upgraded and replaced;

Flexible mission scaling: the number of power units can be increased or decreased at any time according to payload requirements.

4.4 Energy-System Layout and Operating Constraints

4.4.1 Electrical Energy as the Preferred Power Form

This system prefers electrical energy as its power source; the advantages of an electric system:

Controllable steady-state thermal loss, supporting long-duration continuous flight;

Low operating noise, fitting diverse airspace environments;

Fast thrust-regulation response, meeting fine multi-unit coordinated attitude control;

Simple piping systems, easier maintenance than chemical propulsion systems.

Inherent shortcoming: storage batteries are heavy, adding to the total transported mass during climb and partly offsetting the drag-reduction gains. This contradiction is a typical engineering trade-off.

4.4.2 Iterative Engineering Approach to Battery Weight

The trade-off brought by battery weight cannot be resolved into a unique optimum by theoretical deduction alone; it can only be optimized through iterative testing:

Measure climb energy consumption under different energy-storage configurations;

Measure the total electrical energy recovered during descent under the same configuration;

Plot the “battery weight — system net energy consumption” curve;

Select the energy-storage configuration with the lowest net energy consumption as the optimum for that mission.

The optimal storage parameters keep changing as battery energy-density technology evolves; this paper gives no fixed optimal battery capacity value — this parameter is a dynamic engineering optimization target, not a fixed physical constant.

4.4.3 Hard Constraint of the Closed-Loop Energy Operation

The basic constraint for the platform to achieve repeated autonomous takeoffs/landings and a self-sustaining energy cycle: The total electrical energy consumed during climb must not exceed 2/3 of the system's total storage capacity.

The reserved 1/3 of storage serves as safety redundancy, covering the following needs:

Energy for attitude stabilization and course correction under sudden atmospheric turbulence;

Lift-compensation energy after local power-unit failures;

Extra energy when environmental changes lengthen the descent path;

An energy buffer for consecutive takeoff-and-landing cycles.

If climb consumption exceeds 2/3 of total storage, even a theoretically maximal descent energy recovery cannot recharge the system to the safe takeoff standard.

The 2/3 storage ratio is an engineering safety red line. Design goals should minimize the climb energy ratio as far as possible, widening the energy-redundancy margin.

Chapter 5 Theoretical Performance Analysis of the System

5.1 Basic Aerodynamic Formulas

The aerodynamic performance deduction throughout the paper rests on the classical drag equation: Parameter definitions: — aerodynamic drag; — air density; — relative airflow speed; — drag coefficient; — reference frontal area.

5.2 Performance Deduction of the Ascent Phase

During climb the drag panels are retracted and the drag coefficient drops markedly. Reference parameter comparison: Conventional rigid transport platform ; This system's retracted configuration ; Reduction of aerodynamic drag:

Energy-consumption expression for the whole climb: The first term is the increment of gravitational potential energy; the second term is the energy spent overcoming aerodynamic drag over the whole path.

5.3 Performance Deduction of the Descent Phase

During descent the drag panels are fully deployed, and the reaches at most close to 1.0. Instantaneous generating power:

Definition of energy-recovery efficiency:

Scaled-model test data show that under the ideal fully open condition the kinetic-energy recovery efficiency can reach 80%–90%.

5.4 Energy Analysis of Inertial Run-Out Through the Dense Atmosphere

An important capability of the platform: relying on the kinetic energy accumulated during climb, it coasts inertially out of the low dense atmosphere.

Working mechanism: In the low-altitude region the platform continuously spends energy overcoming gravity and aerodynamic drag; as altitude increases,

atmospheric density decays exponentially with height: The atmospheric scale height is about 8.5 km.

Above 20–30 km the air density has decayed drastically and aerodynamic drag is essentially negligible. The platform no longer needs continuous thrust to fight drag, coasting upward on its existing kinetic energy.

At the top of the stratosphere (about 50 km) air density is only about 0.3% of the sea-level value. The platform continues coasting on inertia until near the Kármán line (100 km).

The essence of this process: the kinetic energy stored during climb is continuously converted into gravitational potential energy, with no sustained power input needed in the later stage.

5.5 Criterion for the Closed-Loop Energy Cycle

Combining the energy balance over the whole flight, the net-energy index is defined as: where: — total electrical energy consumed during climb; — total electrical energy recovered during descent.

Criterion: when , the platform possesses a theoretical self-sustaining cyclic operating capability.

Test deduction shows that for normal medium-payload, medium-altitude conditions a working region satisfying exists; when the payload is too large or the climb altitude exceeds the critical value, the case turns into net energy consumption.

Chapter 6 Experimental Verification Plan

6.1 Design of the Two-Tier Test System

The whole verification plan is divided into two tiers: component-level tests and full-system integration tests. Component-level tests:

Individual power-characteristic calibration of the tiltrotor and ducted-rotor units;

Response-speed and structural-strength tests of the drag-panel deploy/retract mechanism;

Calibration of motor windmilling generating power under different simulated wind speeds.

System-integration tests:

Build a small 4–6-unit scaled mesh verification platform;

Measure climb-phase energy consumption and descent-phase total energy recovery;

Deliberately shut down some power units to test the platform's fault tolerance.

6.2 Summary of Key Expected Test Results

Test item	Expected target	Remarks
Climb aerodynamic-drag reduction	40%~60%	Mesh configuration + retracted drag panels
Kinetic-energy recovery efficiency during descent	80%~90%	Drag panels fully deployed
Net-energy state under normal conditions	Positive (generation > climb consumption)	Medium payload, medium climb-altitude band
Net-energy state under extreme conditions	Negative (generation < climb consumption)	Overload and extreme-altitude mission scenarios
Fault-tolerance threshold	Tolerates 10%~20% power-unit failure	Remaining healthy units compensate lift and keep stability

6.3 Engineering Iteration Approach

It must be made clear that the platform's optimal operating parameters are an engineering-iteration optimization problem, not a basic-architecture theory problem. The optimal combination of storage capacity, motor power, panel area and number of networked units depends heavily on the mission scenario. The recommended iteration path:

Build several groups of scaled prototypes with different parameter configurations;

Run grouped ascent-descent cycle tests, collecting energy-consumption and power-generation data;

Compare test results horizontally and select the parameter combination best matching the target mission.

Parameter-optimization iterations do not affect the feasibility of the core architecture; once the basic framework is established, parameter tuning is progressive engineering refinement.

Part I Summary

This paper presents a multi-rotor mesh-spliced lift-transport aircraft system based on bidirectional air-drag harvesting. The main conclusions are summarized below:

Air drag has bidirectional resource-utilization potential. During climb the retraction mechanism suppresses drag and cuts energy consumption; during descent the deployed panels capture airflow kinetic energy for generation, turning air drag from a one-way loss factor into a bidirectional usable resource.

The mesh modular configuration provides high redundancy fault tolerance. Power units are networked in parallel; local module failures do not directly cause loss of control of the whole vehicle; the building-block detachable structure greatly lowers O&M cost and extends the platform's service life.

The system has a unique payload property: the greater the total gravity, the stronger the descent power generation. This property gives the platform an energy advantage over conventional aircraft in heavy-payload transport.

The blade-like lateral gliding re-entry scheme can lower re-entry thermal load. By lengthening the glide path and decelerating gently, the aerodynamic-heating peak is spread out, simplifying the thermal-protection structure required of the spacecraft.

Self-closed energy-loop operation is theoretically feasible. Within suitable payload and altitude ranges, the electricity recovered during descent can cover most of the climb consumption, greatly reducing the need for continuous external energy supply.

The core value of this scheme lies in providing a reusable lift-transport paradigm with no chemical jet propulsion and with energy-cycle potential. The system is not a local improvement of existing aircraft; it offers a new technical route for high-altitude transport and space pre-boosting.

(End of Part I)

Part II: Structural Design, Energy Management, Application Scenarios and System Comparison

7.1 Structural Forms of the Drag Panels

The drag panels are the core actuators of energy recovery during descent. This system adopts two optional drag-panel structural forms: the petal-type deployment structure and the louver-type flipping structure. Each has its own characteristics and can be selected according to the mission scenario.

7.1.1 Petal-Type Deployment Structure

The design of the petal-type deployable drag panel is inspired by the blooming mechanism of flowers in nature. Multiple fan-shaped panels are hinged to the aircraft body and swing outward during descent, forming a large-area drag surface resembling an umbrella or a blossom.

Structural features:

Composed of 8–16 fan-shaped panels arranged ring-wise around the aircraft body

Each panel is independently driven, and its deployment angle can be controlled separately

The panels use lightweight composite materials (carbon fiber or aramid honeycomb structure)

After deployment the overall shape is circular or plum-blossom-like, increasing the frontal area by 3–5 times

When stowed, the panels fold flush against the fuselage, keeping a streamlined shape during ascent

Advantages:

Large drag area after deployment, high air-drag capture efficiency

Even force distribution with a reasonable structural stress layout

Self-stabilizing deployment (the aerodynamic center aligns automatically with the center of mass)

High visual recognizability, convenient for ground observation and remote control

Damage to a single panel does not affect the deployment and operation of the others

Disadvantages:

Still occupies some space when stowed and cannot lie fully flush against the fuselage

Fairly sensitive to crosswinds; in strong-wind environments it may need to narrow its deployment

Relatively complex manufacturing process with many hinges

Requires precise synchronous control to avoid uneven deployment

Applicable scenarios: missions requiring larger generating power, or descents with high trajectory verticality.

7.1.2 Louver-Type Flipping Structure

The louver-type flipping drag panel consists of multiple strip-like blades arranged in parallel, switching between the "open" (generating drag) and "closed" (reducing drag) states through a flipping mechanism.

Structural features:

Composed of 10–20 strip-like blades arranged in parallel

The blades flip in linkage through a rod mechanism

When closed, the blades lie flush against the fuselage surface, forming a streamlined shape

When open, the blades stand perpendicular to the airflow, producing maximum drag

Each blade can independently adjust its flip angle, achieving fine-grained drag control

Advantages:

Flush against the fuselage when stowed, with minimal drag during ascent

Simple structure with high reliability

Blades can open stage by stage, achieving progressive drag regulation

Lower manufacturing and maintenance cost

Gaps between blades let airflow pass, reducing turbulence noise

Disadvantages:

Limited deployed drag area; total drag smaller than the petal type

Airflow leakage between blades slightly lowers energy-capture efficiency

Blade pivot axes need high-durability design (frequent flipping)

In high-speed airflow the blades may vibrate (damping design required)

Applicable scenarios: missions with frequent takeoffs and landings, or with high streamlined-fuselage requirements.

7.1.3 Comparison of the Two Structures and Selection Advice

Property	Petal type	Louver type
Maximum drag area	Large (3-5x frontal area)	Medium (2-3x frontal area)
Energy-recovery efficiency	85%-90%	75%-85%
Streamlining when stowed	Good	Excellent
Structural complexity	Higher	Lower
Manufacturing cost	Higher	Lower
Maintenance difficulty	Medium	Low
Strong-wind adaptability	Fairly sensitive	Fairly stable
Fault tolerance	Damage to a single panel does not affect the whole	A damaged blade can be locked and isolated
Recommended scenario	Deep-space return, heavy-load descent	Regular takeoff/landing, inter-city flight

Recommended scheme: for most mission scenarios, the louver-type flipping structure is preferred for its simple structure, high reliability and low maintenance cost; for deep-space or heavy-payload missions requiring maximum energy recovery, the petal-type deployment structure can be chosen.

7.1.4 Material Selection for the Drag Panels

The drag panels must balance light weight, strength, heat resistance and cost.

Material				Cost	

	Density (g/cm^3)	Specific strength	Heat resistance (°C)		Applicable location
Carbon-fiber composite	1.6	Extremely high	150-200	High	Primary structure, hinges
Aramid honeycomb sandwich	1.4	High	100-150	Medium	Panel body
Glass-fiber composite	1.8	Medium	100-150	Low	Secondary load-bearing parts
Titanium alloy	4.5	Medium	500+	Very high	Hinges in high-temperature zones

Recommended scheme: the panel body uses aramid honeycomb sandwich structure (light, high toughness); hinges and connectors use titanium alloy (high-temperature resistant, fatigue resistant); carbon fiber serves as the main frame material (high strength, low weight).

7.2 Connection Methods of the Mesh-Splicing Structure

The mesh-splicing structure of this system must solve one core engineering problem: how can multiple independent rotor units be firmly connected together while each keeps its operational independence?

7.2.1 Mechanical Connection

A frame-type connection scheme is adopted:

Each rotor unit connects to the main frame through 4–6 connection points

The connection points use quick-release locking mechanisms, supporting rapid installation and removal

The main frame is a lightweight aluminum-alloy or carbon-fiber-composite truss structure

Gaps are reserved between units to avoid mutual airflow interference

The overall structure forms a honeycomb-like or triangular mesh layout, ensuring even force transmission

Core principles of frame design:

Strong enough: to withstand aerodynamic and gravitational loads in flight

As light as possible: not becoming the main source of system weight

Easy to assemble and disassemble: supporting rapid unit replacement

Fatigue-resistant: to endure the cyclic loads of repeated takeoffs and landings

7.2.2 Electrical and Control Connections

Each unit joins the main system through a standardized interface (power lines + communication lines)

The communication protocol adopts a redundant dual-bus architecture (main bus and backup bus)

Hot swapping is supported: failed units can be replaced while the system is running

Each unit is independently powered and controlled, not relying on other units

Power lines use ring redundant wiring: even if one segment breaks, supply continues through other paths

7.2.3 Balancing Connection Strength and Weight

The design of connection points must find a balance between “strong enough” and “as light as possible”

Finite element analysis (FEA) is used to optimize the shape and material thickness of the connection points

Lightweight target: the total weight of the connection structure shall not exceed 10%–15% of the system's total weight

Key connection points adopt a double-insurance design (primary connection + backup connection)

7.2.4 Maintainability and Damage-Resistance Advantages of the Mesh Structure

The multi-rotor mesh-splicing structure differs fundamentally from conventional integrated aircraft in maintenance and survivability:

1. Independent replacement without whole-vehicle repair

When a conventional aircraft's structure is damaged, it often requires extensive disassembly, factory repair, or even scrapping of the whole vehicle. The mesh-splicing structure of this system supports independent module replacement — whenever any rotor unit fails or is damaged, that unit alone is detached from the mesh frame and a new unit is installed to restore full function. The whole repair

involves no other unit and needs no vehicle-level disassembly, greatly reducing maintenance difficulty and time cost.

2. Partial unit failure does not affect overall function

The core advantage of the mesh structure lies in its high redundancy. Even if some units (such as rotors or drag-panel modules) fail or detach during flight, the remaining units can maintain stable flight of the platform by dynamically adjusting thrust and attitude. The system does not fail as a whole due to local damage.

3. Damage resistance in the space environment

In the space environment, aircraft face unpredictable risks such as micrometeoroid and debris impacts. For conventional integrated aircraft, structural damage at any critical part can lead to scrapping the whole vehicle. The mesh-splicing structure disperses the risk across independent units: even if one unit is struck and damaged, it is merely "one block of the building blocks broken", while the overall frame and the remaining units keep working normally. This property gives the system survivability far beyond conventional aircraft in deep-space missions.

4. The economics of "break one, replace one"

The modular design of the mesh structure brings notable full-lifecycle economic advantages:

No need to scrap the whole vehicle

No need for factory overhauls

On-orbit repair or rapid ground replacement is possible

System life is greatly extended, and the amortized cost per mission is extremely low

This "building-block" repair logic keeps the system highly available at extremely low maintenance cost over its full lifecycle.

7.2.5 Minimum Number and Recommended Range of Multi-Unit Splicing

Minimum number: 6 units

Six units are the minimum needed to form a stable mesh structure. In this configuration:

A basic honeycomb or hexagonal layout can be formed

Preliminary redundancy fault tolerance exists (losing 1 unit does not affect the whole; losing 2 can be compensated by adjustment)

The ratio of structural weight to lift capacity remains within an acceptable range

Recommended range: 12–24 units

Within the range of 12–24 units the overall performance of the system is optimal:

Above 12 units: redundancy rises markedly, and 2–3 failed units still allow normal operation

Around 18 units: the best cost-effectiveness, with optimal payload capacity per unit cost

Below 24 units: structural weight remains controllable

Beyond 24 units: diminishing marginal returns

Past 24 units, the extra payload capacity from each additional unit is offset by the increased energy consumption from added structural weight. Unless special mission needs (such as super-heavy payloads) exist, exceeding 24 units is not recommended.

7.3 Layout of the Energy-Recovery Modules

Each rotor unit integrates one energy-recovery module below it, containing:

Generator unit: the motor working in reverse mode, converting mechanical energy into electrical energy

Rectifier circuit: converting alternating current into direct current

Voltage regulator: stabilizing the output voltage to match battery-charging requirements

Energy-management controller: monitoring generating power in real time and dynamically adjusting recovery intensity

Layout principles:

The energy-recovery module of each rotor unit works independently without mutual interference

Energy recovered by multiple units is collected to the main battery pack through a common bus

The common bus uses a redundant design; a single-path failure does not affect overall recovery

The energy-recovery module shares the same windings and magnetic circuit with the power motor, reducing extra weight

Physical basis of bidirectional motor operation:

A motor is both an electric motor and a generator, depending on the relation between current direction and rotation direction.

Forward mode: current → motor rotation → thrust produced (ascent phase)

Reverse mode: airflow drives blade rotation → motor turns backwards → current produced (descent phase)

This "bidirectional operation" capability requires no extra mechanical switching, only a polarity switch of the power-electronics circuit, with millisecond-level response time.

Chapter 8 Energy-Management Strategy

The energy management of this system involves simultaneous charging and discharging of multiple independent units and requires an intelligent allocation and dispatch strategy.

8.1 Energy-Allocation Strategy During Ascent

During ascent, energy is mainly used to drive the rotor units to produce lift.

Allocation principles:

Power is allocated to each unit according to its current state (temperature, load, health)

Units in good condition and high efficiency take on more load

Units in poor condition or overheated are derated as protection

Total power demand is computed centrally by the main controller and executed in a distributed manner

Power allocation is updated every 10 milliseconds to adapt to real-time changes of flight state

Redundant switching logic:

When a unit fails, its load is automatically reassigned to neighboring units

Switching time is less than 50 milliseconds (imperceptible to the human eye)

Overall thrust loss of the system does not exceed 5% (when the number of failed units is ⩽10%)

Load-balancing algorithm:

```
Plain Text IF unit_state = healthy THEN assign standard load IF unit_state = overheated THEN assign 80% standard load IF unit_state = failed THEN assign zero, trigger redundancy switching IF unit_state = recovering THEN assign 60% standard load, recover gradually
```

8.2 Energy-Recovery Strategy During Descent

During descent each rotor unit enters generating mode. The degree of drag-panel deployment and the descent speed jointly determine the recovery power.

Recovery strategy:

Each unit generates, rectifies and regulates independently

The optimal generating load is chosen by unit temperature (full load when cool, derated when hot)

The battery pack's state of charge (SOC) affects recovery priority: charging is prioritized when battery capacity is low

Dynamic balance of recovery power: when the drag panels are fully deployed, the system automatically adjusts the generating load so that the descent speed stays at the preset value

Allocation of recovered power:

Total recovered power is first used to charge the main battery pack

When the main pack nears full charge, recovered energy is diverted to auxiliary storage (supercapacitors or backup batteries)

In emergencies, recovered energy can directly supply other units (such as attitude control and heating systems)

Relation between generating load and descent speed:

The larger the generating load, the greater the braking force and the slower the descent

The slower the descent, the lower the recovery power per unit time ()

Therefore a balance must be found between “as slow as possible” and “maximum generation per unit time”

Optimal descent speed: about 5–10 m/s (depending on the specific configuration)

8.3 Emergency Energy Reserve

The 1/3 redundancy reserve principleapplies at three levels:

Battery reserve:

Depth-of-discharge limit: no more than 80% (20% retained as emergency reserve)

The emergency reserve is never touched unless the system judges that a crash risk exists

Release of the emergency reserve requires manual confirmation (ground console) or an AI decision (under extreme circumstances)

Unit reserve:

One or two rotor units are always kept in “standby” state (low or zero load)

When other units fail, the standby unit immediately joins at full load

Standby duty rotates among units to avoid performance degradation from prolonged idling

Path reserve:

The ascent path plan reserves an “emergency return altitude” — before reaching it, the energy recovered during descent is enough to cover the return

Beyond this altitude the system enters “one-way mode”, landing only at the destination and not returning

Emergency return paths are pre-computed and stored in the flight-control system

8.4 Boundary Conditions of the 1/3 Redundancy Reserve and Emergency Mode

Scope of applicability of the 1/3 redundancy:

The 1/3 redundancy (ascent-phase energy consumption not exceeding 2/3 of the total energy reserve) is computed under the following “standard conditions”:

Payload weight within 70%–100% of the system's design load

Target altitude not exceeding 50 km (upper-middle stratosphere)

Good weather conditions (wind speed below 15 m/s, no strong turbulence)

Health of all units above 90%

Insufficient redundancy under extreme conditions:

When several of the following extreme conditions overlap, the 1/3 redundancy may be insufficient to ensure a safe return:

Payload weight reaches the system limit (100%–120%)

Target altitude exceeds 50 km (more energy must be consumed)

Strong wind or turbulence environment (extra energy needed to maintain attitude)

Degraded health of some units (thermal damage or minor faults)

Emergency mode:

Under the above extreme conditions the system automatically switches to "emergency mode":

Deliberately give up part of the recovery efficiency: reduce drag-panel deployment, prioritizing the safe range of descent speed

Energy first for safe return: recovered electricity is not stored in the battery but supplied directly to attitude control and critical units

Simplify the descent path: choose the shortest descent path (sacrificing generation for shorter loiter time)

Emergency power allocation: concentrate all available energy on critical units, temporarily shutting down non-critical ones (such as some sensors)

Manual-takeover prompt: if the situation remains uncontrollable, the system requests ground-console intervention for decision

Trigger conditions of emergency mode:

The system predicts the remaining energy is insufficient to complete the full descent path (based on integrated computation of current altitude, speed, payload and weather)

Or the main battery pack's SOC (state of charge) falls below the preset safety threshold

Exiting emergency mode:

When the system predicts the remaining energy suffices for a safe return, it automatically switches back to standard mode

Or exits manually after ground-console confirmation

8.5 Multi-Mode Energy Management

The system supports three energy-management modes, switching automatically according to mission needs:

Mode	Applicable scenario	Energy strategy
Economy mode	Short-range lift, regular missions	Recovery first, charging first, energy consumption minimized
Standard mode	Medium distance, regular payload	Balance consumption and recovery, minimize net energy use
High-performance mode	Heavy payload, deep-space return, extreme missions	All energy for mission completion, recovery efficiency secondary

Mode-switching logic:

Based on mission input parameters (target altitude, payload weight, planned return time), the system automatically recommends the optimal mode

Operators can override manually (via the ground console or a preset mission plan)

Mode switching is unrestricted during flight, but flight-state stability must be maintained during the switch

Chapter 9 Application-Scenario Analysis

The application scenarios of this system span from near-Earth transport to deep-space missions, from commercial launches to emergency rescue.

9.1 Independent Space Round-Trip Vehicle

This is the most core and most disruptive application scenario of the system: serving as a fully independent space round-trip vehicle — no dependence on rockets, launch sites, or external fuel resupply.

Complete mission flow:

Ground takeoff: the system lifts off vertically from any flat ground (no dedicated launch site needed)

Ascent phase: rotor units provide lift, drag panels stowed, crossing the dense atmospheric region at minimal drag

Inertial run-out: after reaching the stratopause, accumulated inertia carries it further upward into the low-Earth-orbit region

Spaceflight: at the outer edge of the atmosphere, inertia maintains orbital motion with almost no energy consumption

Mission execution: performs transport, observation, communication-relay and other tasks in low-Earth orbit or suborbital space

Re-entry return: after completing the planned task, enters the designated re-entry window by adjusting attitude and thrust direction

Leaf gliding: drag panels deploy and the platform enters the atmosphere slowly along a lateral path

Energy recovery: continuous generation during descent recharges the battery pack with recovered energy

Landing and reuse: lands safely at the designated site and can be reused after inspection

Comparison with rocket launches:

Property	Rocket launch	Independent round trip by this system
Launch-site requirement	Dedicated launch site required	Any flat ground
Fuel requirement	Hundreds of tons of chemical fuel	Only electrical energy
Return method	Fuel resupply or recovery failure	Natural return by leaf gliding
Reuse rate	Low (partial recovery, cost still high)	High (hundreds of uses)
Mission cost	Tens of millions - hundreds of millions of dollars	Millions - tens of millions of dollars
Mission preparation cycle	Months	Days to weeks

This system offers “a new option beyond rockets” for medium-payload (hundreds of kilograms to a few tons) low-Earth-orbit missions.

9.2 Launch Assistance for Starlink-Type LEO Constellations

The core problem facing current LEO satellite-constellation launches is: each launch is costly, infrequent and inflexible. Even though SpaceX's Falcon 9 has cut launch cost to roughly a few thousand dollars per kilogram, for constellation plans of thousands of satellites the total cost remains extremely high.

The solution of this system:

Deploy the satellite cluster on the lift platform, which carries it to an altitude above the stratosphere (about 30–50 km)

The platform continues rising by inertial coasting, approaching the edge of the atmosphere

The satellites are released at this altitude and start their own small thrusters to complete final orbit insertion

Comparison with rocket launches:

Property	Rocket launch	Launch assistance by this system
Fuel consumption	Extremely large (tens of thousands to hundreds of tons)	Extremely small (electricity + gravity)
Launch cost	Extremely high (tens of millions - hundreds of millions of dollars)	Lower (electricity + maintenance)
Launch frequency	Limited (site, weather, fuel)	Flexible (electric recharge suffices)
Environmental impact	Large (fuel combustion, noise)	Small (electric operation, quiet)
Payload weight limit	Heavy payload (a few tons to tens of tons)	Medium payload (hundreds of kilograms to a few tons)
Launch window	Limited by weather and launch-site scheduling	Launch anytime

9.3 Inter-City Cargo and Passenger Transport

Short range (tens to hundreds of kilometers):

Vertical takeoff and landing, no runway needed

Electrically driven, low noise, suitable for urban environments

Can be deployed on rooftops, parking lots and other limited spaces

Moderate speed, quieter and more economical than helicopters

Noise level: about 65–75 dB (versus 90–100 dB for helicopters)

Medium range (hundreds to over a thousand kilometers):

Tiltrotor mode gives higher cruise efficiency

Point-to-point transport within the atmosphere is possible

Energy cost far below helicopters (electricity vs. aviation fuel)

Lower noise than helicopters, with clearly reduced noise pollution

Economic comparison:

Transport mode	Energy per km per ton	Cost per kg per km
Helicopter	High (fuel)	about $2-5
Electric vertical takeoff and landing aircraft	Medium (battery)	about $1-3
This system	Low (electricity + recovery)	about $0.3-1
Truck / train	Very low	about $0.05-0.2

In short-to-medium transport this system is slightly more expensive than ground transport but faster; in long-range transport it is far below helicopters, approaching the economics of electric vertical takeoff and landing aircraft.

9.4 Leaf-Glide Re-Entry: A Thermal-Protection Breakthrough for Deep-Space Return

The most potentially disruptive application of this system is: serving as a component of a “space round-trip vehicle”.

Flow:

A rocket delivers the payload to low-Earth orbit or beyond

The lift platform takes off from Earth's surface (electrically driven) and climbs to the top of the stratosphere

The platform acts as the “return capsule” of the space vehicle, separating from the rocket to operate independently

After completing the space mission, it returns to Earth's atmosphere

Leaf-glide mode: lateral gliding, slow deceleration, no dependence on a heat shield

Magnetic-drag generation recovers the descent energy

Safe landing, system reuse

Key points:

This is not a copy of “reusable rockets” — the return requires no rocket fuel at all

The descent phase not only consumes no fuel but actually generates power

Extremely high reuse rate (structural life can reach hundreds of missions)

More energy-saving and economical than any existing reusable scheme

9.5 Other Potential Application Scenarios

Scenario	Description	Advantage of this system
Emergency rescue	Material delivery to landslide, flood and earthquake areas	Vertical takeoff and landing, no runway, deployable in extreme environments
High-altitude base station	Carrying communication equipment to provide broadband coverage from the stratosphere	Long loiter, solar charging, multi-platform cooperative networking
Scientific research	Atmospheric sampling, high-altitude astronomical observation, climate research	Fine adjustment of altitude and position, long loiter possible
Military applications	Battlefield reconnaissance, supply delivery, communication relay	Low detectability (electric), high-redundancy survivability
Sightseeing and tourism	High-altitude sightseeing, suborbital experience	Quiet, smooth, reusable

9.6 Priority Ranking of Application Scenarios

Based on technology maturity, market demand and competitive landscape, applications are recommended to advance in the following priority:

Priority	Application scenario	Rationale
1	Emergency rescue and material delivery	Relatively low technical threshold, urgent market demand
2	High-altitude base station and communication relay	Gap in existing technology, little competition
3	Starlink-type satellite launch assistance	Huge market, but more validation needed
4	Inter-city cargo and passenger transport	Needs airspace management and supporting regulations
5	Independent space round-trip vehicle	Technically most complex, yet most disruptive

Chapter 10 Comparison with Existing Schemes

This chapter compares the strengths and weaknesses of this system against existing transport schemes along multiple dimensions.

10.1 Traditional Rockets

Property	Rocket	This system
Thrust source	Chemical fuel combustion	Electric drive (battery + recovery)
Launch cost	Extremely high (tens of millions - hundreds of millions of dollars per launch)	Lower (millions - tens of millions of dollars per launch)
Reuse rate	Low (requires recovery or partial recovery)	High (structure reused hundreds of times)
Maintainability	Poor (factory overhaul)	Good (modular replacement)
Safety redundancy	Low (high single-point-failure risk)	High (mesh redundancy, fault-tolerant design)
Environmental friendliness	Poor (fuel-combustion pollution, huge noise)	Good (electric, low noise)

Launch-site requirement	Dedicated launch site required	Any flat ground
Launch preparation cycle	Months	Days to weeks
Payload weight range	Large (a few tons - tens of tons)	Medium (hundreds of kg - a few tons)

10.2 Helicopters / Tiltrotor Aircraft

Property	Helicopter	This system
Service ceiling	About 6-8 km	Can reach the edge of the atmosphere (about 100 km)
Speed	About 200-300 km/h	Medium speed within the atmosphere, inertial coasting outside
Range	Hundreds of kilometers (refueling needed)	Long-duration flight possible (energy recovery)
Energy consumption	High (aviation fuel)	Low (electricity + recovery)
Fault tolerance	Low (single-rotor failure means falling)	High (multi-unit redundancy)
Noise	Extremely high (90-100 dB)	Lower (65-75 dB)
Maintenance cost	High (complex mechanical structure)	Low (modular replacement)
Vertical takeoff-and-landing capability	Yes	Yes
Runway requirement	Not needed	Not needed

10.3 Large Drones

Property	Drone	This system
Payload capacity	Tens - hundreds of kilograms	Hundreds of kilograms - a few tons
Service ceiling	About 10-15 km	Can break through the atmosphere (about 100 km)

Endurance	Hours (battery-limited)	Longer (extended by energy recovery)
Scalability	Limited (single-aircraft design)	Strong (mesh splicing, units stackable)
Redundancy fault tolerance	Limited (some models have redundancy)	High (full redundancy)
Noise	Medium	Medium-low
Maintenance cost	Medium	Low (modular)

10.4 Overall Comparison Conclusion

This system is not a "replacement" for any existing transport scheme; it fills the gaps between existing schemes:

Cheaper than rockets, flying higher than helicopters

Greater lift capacity than drones, better reusability than rockets

No dependence on fuel, no dependence on runways, no dependence on launch sites

It holds a unique cost advantage in the mission band of medium altitude (0–100 km) and medium payload (hundreds of kilograms to a few tons)

Mission characteristics best suited to this system:

Medium payload: hundreds of kilograms to a few tons (not tens-of-tons class)

Medium altitude: 0–100 km (not deep-space distances)

Vertical takeoff and landing required (no runway)

Multiple reuses required (high economic requirements)

Rapid response required (flexible launch windows)

This system is unsuitable for the following missions:

Super-heavy payloads (above 10 tons)

Ultra-long distances (beyond low-Earth orbit)

Extremely high speeds required (such as missile interception)

Chapter 11 Limitations and Future Directions

11.1 Current Theoretical Limitations

Limitation 1: insufficient power supply at high altitude

Above the stratosphere (>50 km) air density is extremely low and the thrust efficiency of rotor units falls sharply. The platform's high-altitude climbing capability relies on accumulated inertia rather than sustained thrust. For missions requiring "continuous climb to ultra-high altitudes", auxiliary propulsion is still needed.

Coping idea: in the higher-altitude band (>50 km), adopt small electric propulsion or plasma propulsion as an aid, or use a "slingshot" trajectory — accelerate near the ground and finish the remaining climb by inertia.

Limitation 2: limited horizontal range

In pure vertical-takeoff mode, the platform's horizontal mobility is limited. Although tiltrotor units support horizontal cruise, under the mesh-splicing structure the overall horizontal range remains constrained by battery capacity and aerodynamic efficiency.

Coping idea: adopt foldable wings (deployed during horizontal flight), converting the multi-rotor structure to fixed-wing mode during cruise to raise the lift-to-drag ratio; or use a "relay" mode — multiple platforms relay the transport, handing over en route.

Limitation 3: extreme-weather adaptability

Performance under strong wind, turbulence and extreme temperature has yet to be verified. Although the redundant structure can cope with adverse environments to some extent, flight capability in typhoons, blizzards and other extreme weather may be significantly limited.

Coping idea: restrict mission weather conditions (wind-speed and temperature ranges), relaxing them gradually as the system matures; or develop an "anti-wind mode" that adjusts unit layout to reduce wind drag.

Limitation 4: control-algorithm complexity

Dozens of independent units working in coordination place high demands on control algorithms. As the number of units grows, control-system complexity increases exponentially. Whether existing flight-control algorithms can support stable control of a large-scale mesh-spliced structure remains to be verified experimentally.

Coping idea: adopt a "master-slave" control architecture — the master controller handles high-level decisions (path planning, energy management), while each unit independently executes low-level control (attitude stabilization, speed regulation). The master controller does not intervene in the details of each unit, reducing the computational burden.

Limitation 5: the positive-feedback limit of payload weight and energy consumption

Although "the greater the gravity, the more power generated" is an advantage, a boundary exists: when payload weight is too large, ascent energy consumption grows faster than descent generation, making net energy negative. This boundary value must be determined experimentally.

Coping idea: provide a recommended payload range (e.g., "recommended load is 60%–100% of the system's design load"); use beyond this range is not recommended.

11.2 Future Research Directions

Direction 1: high-altitude auxiliary propulsion

Study auxiliary propulsion schemes for the high-altitude band (30–100 km), such as small electric propulsion and plasma propulsion, to strengthen high-altitude climb capability and mission flexibility.

Direction 2: solar-electric hybrid energy system

Integrate thin-film solar cells on the drag-panel surfaces to continuously replenish electrical energy during daytime flight. Solar power and drag recovery form dual energy sources, further extending endurance and mission radius.

Direction 3: intelligent distributed control algorithms

Develop neural-network-based distributed control algorithms to give the mesh-spliced structure "self-organizing" capability — when some units fail, the rest automatically reorganize the lift distribution without master-controller intervention.

Direction 4: scale-up experimental verification

Progressively verify scalability and performance limits from 6 units and 12 units to 18 and 36 units, building a performance database at different scales to provide design bases for engineering application.

Direction 5: cost-benefit model analysis

Build a complete economic model covering manufacturing cost, operating cost, maintenance cost and life cycle, quantifying this system's economic advantage over rockets, helicopters and drones to provide data support for its commercialization.

Direction 6: extreme-environment adaptability research

Through wind-tunnel tests and numerical simulation, study system performance under strong wind, turbulence and extreme temperature, establishing a "safe flight envelope" to support actual operations.

Chapter 12 Experimental Verification

12.1 The Most Economical Experimental Verification Path

Verification of this system need not be done in one step. The following staged verification path is recommended:

Stage 1: small verification platform of 3–4 units

Goal: verify whether the basic premise "consume energy on the way up, generate power on the way down" holds

Splice 3–4 rotor units into the minimal mesh structure

Conduct takeoff-and-landing tests at 10–50 m altitude

Measure ascent-phase energy consumption and descent-phase power generation

If generation exceeds 40% of consumption (preliminary target), the principle is established

Cost: about $5,000–20,000 (depending on the choice of rotor units)

Stage 2: extended verification platform of 6–8 units

Goal: verify fault tolerance, redundancy switching and drag-panel effectiveness

Repeat the tests at the 6–8-unit scale

Simulate partial unit failures and observe the system response

Test the effect of drag-panel deployment/retraction on power generation

Cost: about $20,000–50,000

Stage 3: functional verification platform of 12–18 units

Goal: verify the performance and stability of the large-scale system

Conduct takeoff-and-landing tests at the target payload (hundreds-of-kilograms class)

Test performance under different altitudes and weather conditions

Collect complete performance data and optimize parameter configuration

Cost: about $100,000–300,000

Stage 4: full-scale prototype

Goal: verify the complete function of the final design

Finish full-profile tests at the target payload and altitude

Run multiple takeoff-and-landing cycles to verify reusability and durability

Cost: about $500,000–2,000,000

Verification time estimate:

Stage 1: 1–2 months

Stage 2: 2–3 months

Stage 3: 3–6 months

Stage 4: 6–12 months

Total: about 12–24 months

12.2 Unit Tests

Unit tests verify the basic performance of individual components:

Test the thrust and energy consumption of a single rotor unit (tiltrotor and ducted rotor)

Test the deployment/retraction response time of the drag panels (target: <0.5 s)

Test the output power of the generating module at different wind speeds (target: efficiency >70% at 8–15 m/s wind speed)

Test the cycle life of the battery pack at different depths of discharge

12.3 System Tests

System tests verify overall performance when multiple units work in coordination:

Test basic lift functions on the small 3–4-unit platform

Test ascent-phase energy consumption and descent-phase generation on the medium 6–8-unit platform

Test fault tolerance under partial unit failure on the large 12–18-unit platform

Test the effect of different drag-panel deployment degrees on power generation

12.4 Key Experimental Metrics

Test item	Expected result	Validation criterion
Air-drag reduction rate during ascent	40%-60%	Compared with conventional designs
Energy-recovery rate during descent	80%-90%	Compared with theoretical calculation
Net energy under typical conditions	Positive (generation > consumption)	Average over 10 consecutive takeoff-landing cycles
Fault tolerance to failed units	Tolerates 10%-20% unit failure	Active unit-cutoff test
Drag-panel response time	<0.5 s	Measured with a high-speed camera
Overall system reliability	>99% (single mission)	Statistical test data
Structural fatigue life	>500 takeoffs and landings	Accelerated fatigue testing

12.5 Recording and Analysis of Experimental Data

Data items to be recorded:

Ascent phase: initial altitude, target altitude, total duration, total energy consumed (Wh), power allocation of each unit, ambient temperature and humidity, wind speed

Descent phase: initial altitude, descent speed, total duration, total generation (Wh), generating power of each unit, battery SOC changes

Environmental parameters: temperature, humidity, wind speed, wind direction, air pressure

System state: temperature of each unit, vibration data (accelerometer), attitude angles

Data-analysis methods:

Compute net energy = generation − consumption (a positive value means net gain)

Compute energy-recovery ratio = generation / descent gravitational potential energy

Plot the "payload weight vs. net energy" curve to find the optimal payload range

Plot the "descent speed vs. generating power" curve to verify the relation

Count unit-failure frequency to verify system reliability metrics

12.6 Engineering-Iteration Logic of Experimental Results

The purpose of experiments is not one-shot verification but discovering what needs adjustment, adjusting, and verifying again.

Iteration loop:

Design experiment → 2. Obtain data → 3. Identify problems → 4. Adjust design → 5. Re-experiment

Common problems and adjustment directions:

Experimental finding	Possible adjustment direction
Ascent consumes too much energy	Add rotor units, reduce weight, optimize the ascent speed profile
Descent generates too little power	Enlarge drag-panel area, optimize descent speed, adjust generating load
Frequent unit failures	Improve heat dissipation, refine control algorithms, reduce load
Excessive structural vibration	Add damping, optimize frame stiffness, match rotor speeds
Net energy negative	Lower the target altitude, reduce payload, enlarge drag panels

The essence of engineering optimization: The architecture proposed in this paper is fixed, but the parameters (unit count, panel area, battery capacity, motor power) are all adjustable. The value of experiments lies in: real data tells you "what this parameter should be tuned to".

Chapter 13 Summary

13.1 Review of Core Contributions

Proposed the transport paradigm of bidirectional air-drag harvesting. Air drag is turned from "an enemy that must be overcome" into "a resource that can be used".

Designed the multi-rotor mesh-splicing structure. Through the splicing of multiple independent units, high redundancy fault tolerance and "building-block" maintainability are achieved.

Demonstrated the feasibility of the closed-loop energy cycle. Ascent energy saved + descent generation ⩾ ascent consumption, giving the system self-sustaining capability.

Proposed the leaf-glide re-entry strategy. A lateral gliding path replaces vertical falling, allowing safe atmospheric re-entry without a heat shield.

Established the positive-feedback relation between gravity load and power generation. Overturning the traditional view that "weight is a burden", it builds the new perspective that "weight is an energy source".

Set the 2/3 energy red-line engineering constraint. Placing "being able to come back" above "being able to go up", ensuring the system is reusable under all operating conditions.

13.2 Technology Readiness Assessment

Based on the current theoretical analysis and preliminary estimates, the technology readiness level (TRL) of this system is about 3–4 (concept-validation stage).

TRL level	Description	Status of this system
1	Basic principles observed	✅ Completed
2	Conceptual design	✅ Completed
3	Experimental validation of key functions	⏳ Pending (small verification platform)
4	Prototype validation in laboratory environment	⏳ Pending (medium verification platform)
5	Prototype validation in relevant environment	⏳ Pending (large verification platform)

6	Engineering prototype demonstration	⏳ Pending
7	Flight demonstration	⏳ Pending
8	Actual system completed	⏳ Pending
9	Mission-proven in actual tasks	⏳ Pending

13.3 Final Positioning

This system is not an improvement of existing aircraft but a new transport paradigm. It depends on no fuel, no runway, no launch site, no single critical component, and no external energy supply.

It treats the atmosphere as a "springboard" and "energy-exchange medium", gravity as an "energy source", and air drag as a "resource" rather than an "obstacle".

It is both a lift platform and a reusable transport system, holding unique cost and efficiency advantages in a specific mission band (medium altitude, medium payload, high-frequency takeoff and landing).

13.4 Conclusion

This study presents a lift-platform aircraft system that does not rely on traditional fuel propulsion, achieving a closed-loop energy cycle and high system redundancy fault tolerance through the design of bidirectional air-drag harvesting and a multi-rotor mesh-splicing structure.

In theory the system features energy saving on ascent, power generation on descent, immunity to partial unit failures, and safe leaf-glide re-entry, and can serve in scenarios such as Starlink launch assistance, inter-city transport, and deep-space round trips.

Follow-up research will verify actual performance through scale-up experiments and optimize parameter configuration through engineering iteration, moving the system from conceptual design toward engineering application.

The ultimate goal of this system is:

To make flight no longer dependent on burning fuel, but on the intelligent use of gravity and air drag.

To make transport no longer dependent on launch-site facilities, but on the self-assembly capability of modular splicing structures.

To make space round trips no longer disposable consumables, but a reusable, self-sustaining, routine mode of transportation.

It is a lift platform.

But its significance goes far beyond a “platform”.